

6. Energy-based Model

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MTH345: Deep Generative Models: A Statistical Perspective



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Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

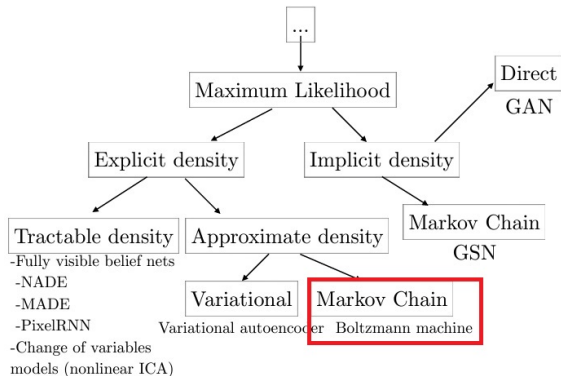
M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

- ▶ In the previous chapters, we reviewed Independent Component Analysis (ICA) and autoregressive models.
- ▶ ICA enjoys fast data generation and dimensionality reduction capabilities, but it may lack the ability to express complex data-generating processes.
- ▶ In contrast, autoregressive models can flexibly model complex joint structures, but their sequential generation process is time-consuming.
- ▶ In this lecture, we will review *energy-based models*, another class of latent variable-based generative models.
- ▶ Under certain conditions, these models can universally approximate arbitrary distributions and offer faster data generation than autoregressive models.

Lecture Topics



► Recommended reference materials include:

- Hinton (2002), “Training products of experts by minimizing contrastive divergence,” *Neural computation*.
- Goodfellow (2016), “Nips 2016 tutorial: Generative adversarial networks,” *arXiv*.

The figure is edited from Goodfellow (2016).

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Limitations of Bayesian Networks



The images are from the Street View House Numbers (SVHN) dataset (Netzer et al., 2011).

Limitations of Bayesian Networks



- ▶ Let us assume, as part of the historical background, that modern houses have progressively transitioned to using blue-backgrounded signs.
- ▶ Let Z_1 indicate whether the house has an administrative origin (e.g., conforming to modern building codes), and Z_2 indicate whether the digit is the ending one.
- ▶ Given $\mathbf{x}$, Z_1 and Z_2 naturally preserve (conditional) independence, i.e.,

$$Z_1 \perp\!\!\!\perp Z_2 \mid \mathbf{x}. \quad (1)$$

Limitations of Bayesian Networks

- ▶ Many latent variable models, including ICA, are formulated as Bayesian networks that satisfy

$$f(\mathbf{z}) = \prod_{j=1}^r f(z_j) \quad \text{and} \quad f_{\theta}(\mathbf{x} \mid \mathbf{z}) = \prod_{j=1}^d f_{\theta}(x_j \mid \mathbf{z}). \quad (2)$$

- ▶ Although the prior latent components $\mathbf{z}$ are statistically independent, after conditioning on the observed data $\mathbf{x}$, the posterior ones can become dependent:

$$f_{\theta}(\mathbf{z} \mid \mathbf{x}) = \frac{\prod_{j=1}^d f_{\theta}(x_j \mid \mathbf{z}) \prod_{j=1}^r f(z_j)}{f_{\theta}(\mathbf{x})} \neq \prod_{j=1}^r f_{\theta}(z_j \mid \mathbf{x}) \quad (3)$$

This phenomenon is known as *explaining away* in directed graphical models.

Energy-based Models

- ▶ Restricted Boltzmann machines (RBMs) are latent variable models that guarantee conditional independence among variables within the same layer.
- ▶ We first review energy-based models and general Boltzmann machines.
- ▶ *Energy-based models* are generative models that utilize an energy function to define a *Boltzmann distribution*¹:

$$f_{\theta}(\mathbf{x}) := \frac{\exp(-E_{\theta}(\mathbf{x}))}{C(\theta)} \quad (4)$$

where $E_{\theta}(\mathbf{x})$ denotes the energy function, and $C(\theta) := \int \exp(-E_{\theta}(\mathbf{x}))d\mathbf{x}$ is the partition function (or normalization constant).

¹Here, we assume unit temperature and a unit Boltzmann constant.

Energy-based Models

- ▶ For example, the multivariate Gaussian distribution with parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is given by

$$f_{\theta}(\mathbf{x}) = (2\pi)^{-d/2} |\boldsymbol{\Sigma}|^{-1/2} \exp \left(-(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) / 2 \right), \quad (5)$$

which yields the energy function

$$E_{\theta}(\mathbf{x}) = \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}). \quad (6)$$

- ▶ Its corresponding partition function (or normalization constant) is $C(\theta) = (2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}$.²

²In general, any distribution in the exponential family, represented by $p_{\theta}(\mathbf{x}) = h(\mathbf{x}) \exp(\boldsymbol{\eta}(\theta)^T \mathbf{T}(\mathbf{x}) - A(\theta))$, can be interpreted as an instance of Equation (4). In this case, the energy function is $E_{\theta}(\mathbf{x}) = -\boldsymbol{\eta}(\theta)^T \mathbf{T}(\mathbf{x}) - \log h(\mathbf{x})$ and the partition function is $C(\theta) = \exp(A(\theta))$.

Boltzmann Machine

- ▶ Boltzmann machines (Ackley et al., 1985) are classical energy-based models that introduce latent variables to define a joint energy density function over both the latent variables and observations.
- ▶ The energy function of a Boltzmann machine is expressed as

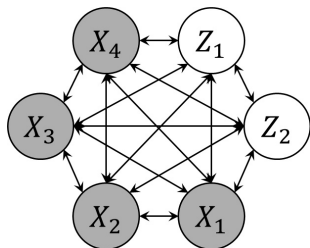
$$E_{\theta}(\mathbf{z}, \mathbf{x}) := - \left(\frac{1}{2} \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix}^T \mathbf{W} \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{pmatrix}^T \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix} \right), \quad (7)$$

where $\theta := (\mathbf{W}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ represents the parameters. Here, $\mathbf{W}$ is a symmetric matrix with zero diagonal elements, indicating the absence of self-loops.

- ▶ The corresponding joint density is then given by

$$p_{\theta}(\mathbf{z}, \mathbf{x}) \propto \exp \left(\frac{1}{2} \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix}^T \begin{pmatrix} \mathbf{W}_{ZZ} & \mathbf{W}_{ZX} \\ \mathbf{W}_{ZX}^T & \mathbf{W}_{XX} \end{pmatrix} \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{pmatrix}^T \begin{pmatrix} \mathbf{z} \\ \mathbf{x} \end{pmatrix} \right). \quad (8)$$

Boltzmann Machine



- For example, when we have $\mathbf{x} \in \{0, 1\}^{d=4}$ and $\mathbf{z} \in \{0, 1\}^{r=2}$, the joint density function can be explicitly expanded as:

$$\begin{aligned}
 p_{\theta}(\mathbf{z}, \mathbf{x}) \propto \exp \bigg(& W_{ZZ,1,2} z_1 z_2 + \sum_{i=1}^2 \sum_{j=1}^4 W_{ZX,i,j} z_i x_j \\
 & + \sum_{j < k} W_{XX,j,k} x_j x_k + \sum_{i=1}^2 \alpha_i z_i + \sum_{j=1}^4 \beta_j x_j \bigg).
 \end{aligned} \tag{9}$$

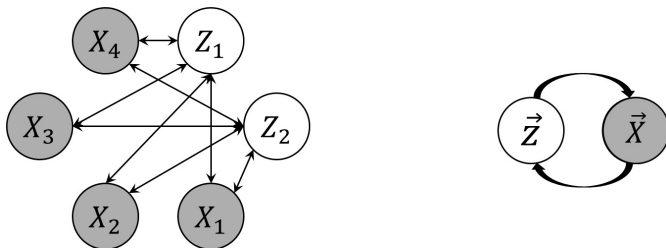
Boltzmann Machine

- ▶ Boltzmann machines assume binary latent variables and binary observed features.
- ▶ The tractable joint energy forms provide an explicit interpretation for logit functions. For a given $\mathbf{x}_{-j} := (x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_d)^T$, let $\mathbf{x}_{j,\text{on}} := (x_1, \dots, x_j = 1, \dots, x_d)^T$ and $\mathbf{x}_{j,\text{off}} := (x_1, \dots, x_j = 0, \dots, x_d)^T$. We then have

$$\begin{aligned} \log \left(\frac{p_{\theta}(\mathbf{z}, \mathbf{x}_{j,\text{on}})}{p_{\theta}(\mathbf{z}, \mathbf{x}_{j,\text{off}})} \right) &= -E_{\theta}(\mathbf{z}, \mathbf{x}_{j,\text{on}}) + E_{\theta}(\mathbf{z}, \mathbf{x}_{j,\text{off}}) \\ &= (\mathbf{W}_{ZX}^T \mathbf{z} + \mathbf{W}_{XX} \mathbf{x} + \beta)_j. \end{aligned} \quad (10)$$

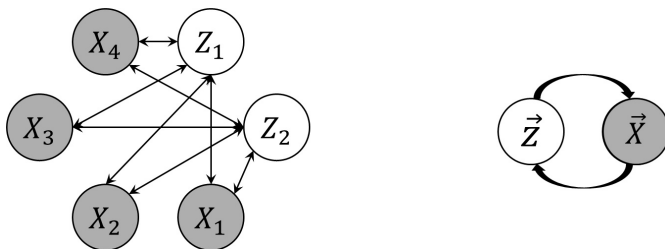
- ▶ However, due to the intra-layer connections within $\mathbf{z}$ and $\mathbf{x}$, represented respectively by $\mathbf{W}_{ZZ}$ and $\mathbf{W}_{XX}$, it remains computationally intractable to evaluate the conditional posteriors $p_{\theta}(\mathbf{z} | \mathbf{x})$, $p_{\theta}(\mathbf{x} | \mathbf{z})$, and their corresponding logits.

Restricted Boltzmann Machine



- ▶ Restricted Boltzmann machines (RBMs) simplify general Boltzmann machines by eliminating connections within latent variables and within observed features, thereby defining a bipartite graph.
- ▶ They were originally proposed by Smolensky et al. (1986) under the name *Harmonium* and gained widespread popularity after Hinton (2002) introduced efficient training algorithms.

Restricted Boltzmann Machine



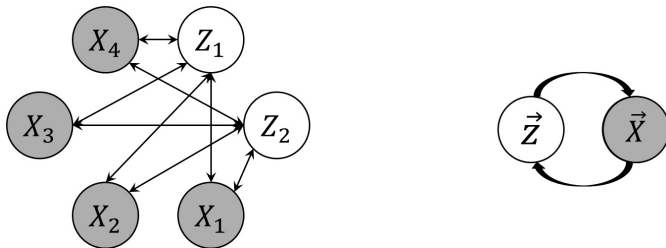
- The energy function of an RBM is explicitly expressed as

$$E_{\theta}(\mathbf{z}, \mathbf{x}) := -(\mathbf{z}^T \mathbf{W} \mathbf{x} + \boldsymbol{\alpha}^T \mathbf{z} + \boldsymbol{\beta}^T \mathbf{x}), \quad (11)$$

which directly corresponds to Equation (8) under the structural constraints $\mathbf{W}_{ZZ} = \mathbf{O}$, $\mathbf{W}_{XX} = \mathbf{O}$, and $\mathbf{W}_{ZX} = \mathbf{W}$.³

³RBM's traditionally assume binary observations and binary latent variables, but they can be extended to generic exponential family distributions (Welling et al., 2004).

Restricted Boltzmann Machine



- For example, when we have $\mathbf{x} \in \{0, 1\}^{d=4}$ and $\mathbf{z} \in \{0, 1\}^{r=2}$, the joint density $p_{\theta}(\mathbf{z}, \mathbf{x})$ simplifies to:

$$p_{\theta}(\mathbf{z}, \mathbf{x}) \propto \exp \left(\sum_{i=1}^2 \sum_{j=1}^4 W_{i,j} z_i x_j + \sum_{i=1}^2 \alpha_i z_i + \sum_{j=1}^4 \beta_j x_j \right). \quad (12)$$

Restricted Boltzmann Machine

- ▶ The joint density $p_{\theta}(\mathbf{z}, \mathbf{x})$ exhibits an exact conditional independence property:

$$p_{\theta}(\mathbf{x} | \mathbf{z}) = \prod_{j=1}^d p_{\theta}(x_j | \mathbf{z}) \quad \text{and} \quad p_{\theta}(\mathbf{z} | \mathbf{x}) = \prod_{i=1}^r p_{\theta}(z_i | \mathbf{x}). \quad (13)$$

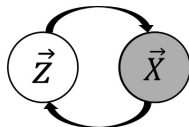
- ▶ Furthermore, RBMs provide analytically tractable inference for each node:

$$p_{\theta}(x_j = 1 | \mathbf{z}) = \text{sig}((\mathbf{W}^T \mathbf{z} + \beta)_j) \quad \text{and} \quad p_{\theta}(z_i = 1 | \mathbf{x}) = \text{sig}((\mathbf{W} \mathbf{x} + \alpha)_i). \quad (14)$$

- ▶ Consequently, RBMs have linear conditional logit functions:

$$\log \left(\frac{p_{\theta}(z_1, \dots, z_i = 1, \dots, z_r | \mathbf{x})}{p_{\theta}(z_1, \dots, z_i = 0, \dots, z_r | \mathbf{x})} \right) = (\mathbf{W} \mathbf{x} + \alpha)_i. \quad (15)$$

Restricted Boltzmann Machine



- RBMs generate synthetic data via the following M -step Gibbs sampling process:
1. Initialize by sampling $\tilde{\mathbf{X}}_1$ from a pre-specified distribution $p(\tilde{\mathbf{x}})^4$ and draw $\tilde{\mathbf{Z}}_1 \sim p_\theta(\mathbf{z} \mid \mathbf{x} = \tilde{\mathbf{X}}_1)$.
 2. For $m = 1, \dots, M - 1$, sequentially generate $\tilde{\mathbf{X}}_{m+1} \sim p_\theta(\mathbf{x} \mid \mathbf{z} = \tilde{\mathbf{Z}}_m)$ and $\tilde{\mathbf{Z}}_{m+1} \sim p_\theta(\mathbf{z} \mid \mathbf{x} = \tilde{\mathbf{X}}_{m+1})$.
 3. The resulting pair $(\tilde{\mathbf{Z}}_M, \tilde{\mathbf{X}}_M)$ constitutes the M -step Gibbs sample. Under certain regularity conditions, the joint density of $(\tilde{\mathbf{Z}}_M, \tilde{\mathbf{X}}_M)$ converges to the true joint distribution $p_\theta(\mathbf{z}, \mathbf{x})$ as $M \rightarrow \infty$.

⁴A common choice is to utilize the empirical distribution of real observations $\mathbf{X}$.

Contrastive Divergence

- ▶ (Restricted) Boltzmann machines aim to find maximum likelihood estimators. However, the log-likelihood function,

$$\log p_{\theta}(\mathbf{x}) = -\log C(\theta) + \log \sum_{\mathbf{z}} \exp(-E_{\theta}(\mathbf{z}, \mathbf{x})), \quad (16)$$

contains the intractable $C(\theta)$, posing a severe computational challenge.

- ▶ Contrastive divergence (CD) (Hinton, 2002) circumvents this issue by evaluating the gradient of the log-likelihood (the score function) directly:

$$\nabla_{\theta} \log p_{\theta}(\mathbf{x}) = - \underbrace{\sum_{\mathbf{z}} \left(\nabla_{\theta} E_{\theta}(\mathbf{z}, \mathbf{x}) \right) p_{\theta}(\mathbf{z} | \mathbf{x})}_{\text{Positive gradient}} + \underbrace{\sum_{\mathbf{x}, \mathbf{z}} \left(\nabla_{\theta} E_{\theta}(\mathbf{z}, \mathbf{x}) \right) p_{\theta}(\mathbf{z}, \mathbf{x})}_{\text{Negative gradient}}. \quad (17)$$

- ▶ The score function is exactly the contrast between the positive and negative phases, which is the direct origin of the name *contrastive divergence*.

Contrastive Divergence

- ▶ Driven by this formulation, the CD algorithm applies short-run Gibbs sampling to approximate the intractable expectations in both phases.
- ▶ For a given training dataset $(\mathbf{x}_i)_{i=1}^n$, the execution of the CD algorithm is formalized as follows:
 1. **Positive Phase:** For a given training observation $\mathbf{x}_i$, sample $\tilde{\mathbf{z}}_i \sim p_{\theta}(\mathbf{z} \mid \mathbf{x} = \mathbf{x}_i)$ to construct the i -th *positive sample* $(\tilde{\mathbf{z}}_i, \mathbf{x}_i)$.
 2. **Negative Phase:** Starting from $\mathbf{x}_i$, draw the i -th *negative sample* $(\tilde{\mathbf{z}}_{i,M}, \tilde{\mathbf{x}}_{i,M})$ using M -step Gibbs sampling.
 3. **Gradient Evaluation:** Compute the empirical positive and negative gradients using the sampled mini-batches.
 4. **Parameter Update:** Update θ by ascending the approximated log-likelihood gradient.

Generation Result

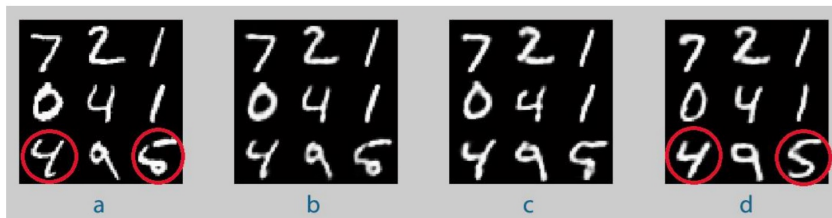


Figure 8: Results from an RBM with 250 hidden units that has been trained on MNIST dataset. In this experiment, after extracting 250 features from 9 MNIST images (28*28 pixel), the new images have been generated from extracted features. (a) 9 MNIST images. (b) Generated images from extracted features with $k = 1$ sampling iteration. (c) Generated images from extracted features with $k = 10$ sampling iterations. (d) Generated images from extracted features with $k = 100$ sampling iterations. The related code is in “*test_generateData.m*”.

Images are from Keyvanrad and Homayounpour (2014).

Flexibility vs. Interpretability

- ▶ Compared with general Boltzmann machines, RBMs sacrifice intra-layer flexibility by eliminating connections within $\mathbf{z}$ and within $\mathbf{x}$.
- ▶ Nonetheless, RBMs function as distributional universal approximators:

Theorem (Representational Power of RBMs; Le Roux and Bengio, 2008)

For any $\epsilon > 0$ and any target probability density function $p(\mathbf{x})$ supported on $\{0, 1\}^d$, there exists an RBM parameterized by θ such that

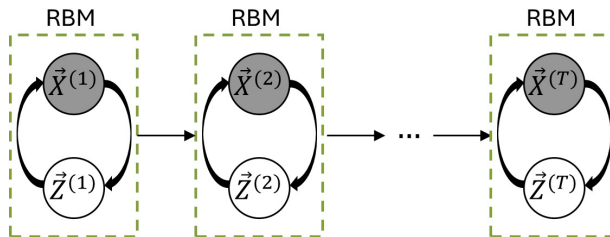
$$D_{\text{KL}}(p(\mathbf{x}) \parallel p_{\theta}(\mathbf{x})) < \epsilon. \quad (18)$$

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1. Restricted Boltzmann Machine

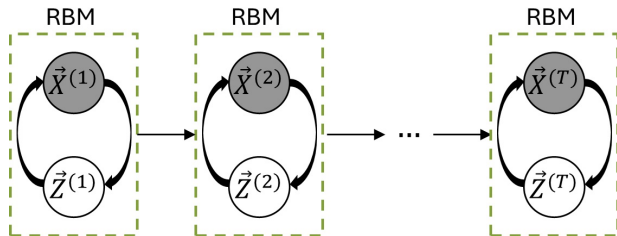
2. Temporal Restricted Boltzmann Machine

Temporal Restricted Boltzmann Machine



- ▶ RBMs face challenges in handling temporal data because they do not permit intra-layer or inter-temporal connections among observations across different time steps.
- ▶ Temporal RBMs (TRBMs) (Taylor et al., 2006) are a temporal extension of RBMs that incorporates directed connections from past latent and observed variables to present ones.

Temporal Restricted Boltzmann Machine

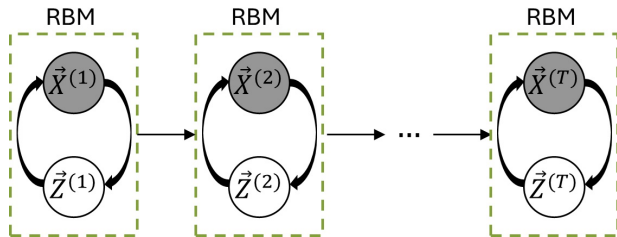


- ▶ TRBMs first construct a standard RBM for the initial time step:

$$p_{\theta}(\mathbf{z}^{(1)}, \mathbf{x}^{(1)}) \propto \exp(-E_{\theta}^{(1)}(\mathbf{z}^{(1)}, \mathbf{x}^{(1)})), \quad (19)$$

where $E_{\theta}^{(1)}(\mathbf{z}^{(1)}, \mathbf{x}^{(1)}) := -((\mathbf{z}^{(1)})^T \mathbf{W} \mathbf{x}^{(1)} + \boldsymbol{\alpha}^T \mathbf{z}^{(1)} + \boldsymbol{\beta}^T \mathbf{x}^{(1)})$.

Temporal Restricted Boltzmann Machine



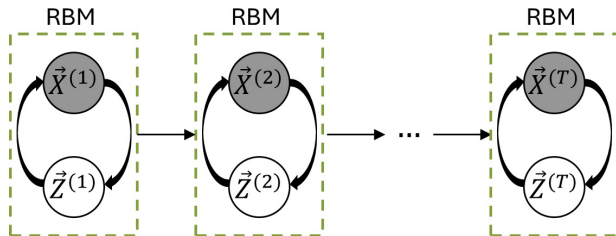
- Next, the conditional distribution of $(\mathbf{z}^{(2)}, \mathbf{x}^{(2)})$ given the initial variables at the first time step can be expressed as:

$$p_{\theta}(\mathbf{z}^{(2)}, \mathbf{x}^{(2)} \mid \mathbf{z}^{(1)}, \mathbf{x}^{(1)}) \propto \exp(-E_{\theta}^{(2)}(\mathbf{z}^{(2)}, \mathbf{x}^{(2)}; \mathbf{z}^{(1)}, \mathbf{x}^{(1)})), \quad (20)$$

where

$$E_{\theta}^{(2)}(\mathbf{z}^{(2)}, \mathbf{x}^{(2)}; \mathbf{z}^{(1)}, \mathbf{x}^{(1)}) := -((\mathbf{z}^{(2)})^T \mathbf{W} \mathbf{x}^{(2)} + (\alpha + \mathbf{U} \mathbf{z}^{(1)})^T \mathbf{z}^{(2)} + \beta^T \mathbf{x}^{(2)}).$$

Temporal Restricted Boltzmann Machine



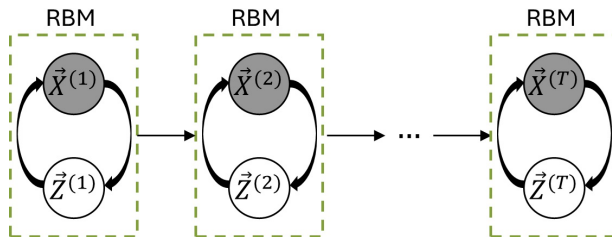
- Finally, TRBMs assume time homogeneity and sequentially stack RBMs:

$$p_{\theta}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:(t-1))}) \propto \exp \left(-E_{\theta}^{(t)}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)}; \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:(t-1))}) \right) \quad (21)$$

where the energy function is defined as

$$E_{\theta}^{(t)}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)}; \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:(t-1))}) := -((\mathbf{z}^{(t)})^T \mathbf{W} \mathbf{x}^{(t)} + (\alpha + \mathbf{U} \mathbf{z}^{(t-1)})^T \mathbf{z}^{(t)} + \beta^T \mathbf{x}^{(t)}). \quad (22)$$

Temporal Restricted Boltzmann Machine



- ▶ Note that $E_{\theta}^{(t)}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)}; \mathbf{z}^{(1:t-1)}, \mathbf{x}^{(1:t-1)})$ is a function of $(\mathbf{z}^{(t-1)}, \mathbf{z}^{(t)}, \mathbf{x}^{(t)})$. Consequently, $p_{\theta}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)} | \mathbf{z}^{(1:t-1)}, \mathbf{x}^{(1:t-1)}) = p_{\theta}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)} | \mathbf{z}^{(t-1)})$.
- ▶ TRBMs function as distributional universal approximators (Odenne and Edwards, 2016).

Temporal Restricted Boltzmann Machine

- ▶ Since $p_{\theta}(\mathbf{z}^{(t)}, \mathbf{x}^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:(t-1))})$ is formulated as an RBM under the energy function $E_{\theta}^{(t)}$, it inherits the exact conditional independence property:

$$p_{\theta}(\mathbf{x}^{(t)} \mid \mathbf{z}^{(1:t)}, \mathbf{x}^{(1:(t-1))}) = \prod_{j=1}^d p_{\theta}(x_j^{(t)} \mid \mathbf{z}^{(1:t)}, \mathbf{x}^{(1:(t-1))}) \quad (23)$$

and

$$p_{\theta}(\mathbf{z}^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)}) = \prod_{i=1}^r p_{\theta}(z_i^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)}). \quad (24)$$

Temporal Restricted Boltzmann Machine

- ▶ TRBMs also provide analytically tractable and interpretable inference for each node:

$$p_{\theta}(z_i^{(t)} = 1 \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)}) = \text{sig}((\mathbf{W}\mathbf{x}^{(t)} + \alpha + \mathbf{U}\mathbf{z}^{(t-1)})_i). \quad (25)$$

Similarly,

$$p_{\theta}(x_j^{(t)} = 1 \mid \mathbf{z}^{(1:t)}, \mathbf{x}^{(1:(t-1))}) = \text{sig}((\mathbf{W}^T \mathbf{z}^{(t)} + \beta)_j). \quad (26)$$

- ▶ Their corresponding conditional logit functions are strictly linear and can be expressed as:

$$\log \left(\frac{p_{\theta}(z_1^{(t)}, \dots, z_i^{(t)} = 1, \dots, z_r^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)})}{p_{\theta}(z_1^{(t)}, \dots, z_i^{(t)} = 0, \dots, z_r^{(t)} \mid \mathbf{z}^{(1:(t-1))}, \mathbf{x}^{(1:t)})} \right) = (\mathbf{W}\mathbf{x}^{(t)} + \alpha + \mathbf{U}\mathbf{z}^{(t-1)})_i. \quad (27)$$

Temporal Restricted Boltzmann Machine

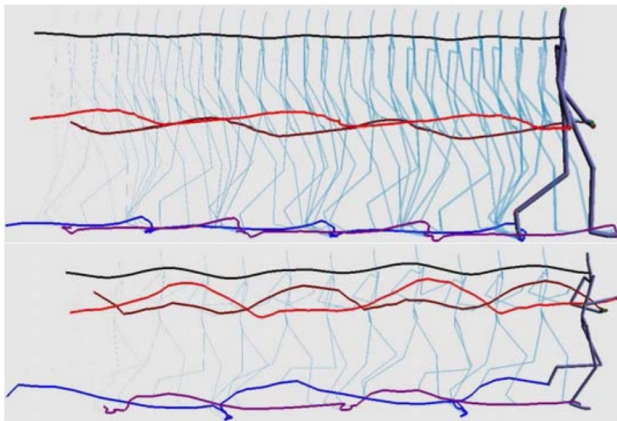
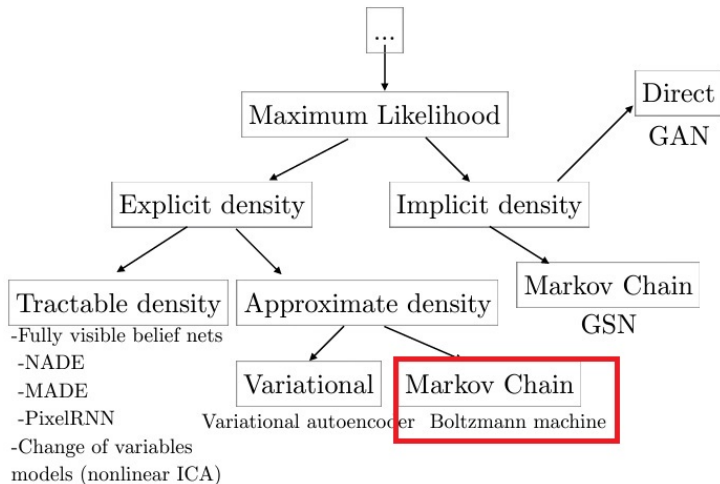


Figure 3: After training, the same model can generate walking (top) and running (bottom) motion (see videos on the website). Each skeleton is 4 frames apart.

Images are from Taylor et al. (2006).

Lecture Topics



The figure is edited from Goodfellow (2016).

References II

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